1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L17 – Case Study on Prototype (contd)**

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3 **Recap - Calendar App Case**

- A calendar app – introduced earlier
  - Recap - we are interested to build a calendar app. It is meant for the students (mainly) to help in their various academic activities

4 **SRS – Recap**

5

Prototype with Heuristic Evaluation

6 **Overview**

- A medium fidelity, vertical prototype
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- Prototype built with figma

7 **Overview**

- Prototype has
  - Month view
  - Specific date view
  - Year view
  - Background view
    - Dark background option
    - Custom Background from device option
  - Views
    - Deadline view
    - Timetable view
    - Schedule view

8 **Year View**

9 **Month View**

10 **Specific Date View**

11 **Set Event View**

12 **Background View**

13 **Deadline View**

14 **Timetable View**

15 **Schedule View**

16 **Prompt for Error Prevention**

17

Heuristic Evaluation - Designer's View

18 **Heuristic Evaluation**

- Used Jacob Nielson's 10 heuristics
  - Visibility of system status
  - Match between system and real world
  - User control and freedom
  - Consistency and standards
  - Error prevention
  - Recognition rather than recall
  - Flexibility and efficiency of use
  - Aesthetic and minimalistic design
  - Help users recognize, diagnose, and recover from errors
  - Help and documentation

19 **Heuristic Evaluation**

- Visibility of system status - system should keep users informed about what is going on, through appropriate feedback within a reasonable time gap.
  - Users can distinguish and easily find today's date and month as it is highlighted with noticeable color.
  - Dates with deadlines are also highlighted in red color.
  - Changing the months is quite intuitive as dragging leftwards takes to the previous month and dragging rightwards takes to the next month, and also on changing the month we can see the change in the view in real-time.
  - While selecting the year, the up-down buttons increase visibility and give users feedback on the change of the year.
  - New events for any particular date can be just created by tapping on the date to which event has to be added.
  - Moreover, in the specific date view for any date, the events that have some deadline is when created has a star symbol which helps to see the user the change taken place.
  - In timetable view, the user can see the current time in a 'red line' thus helping in an easy preview of current events

20 **Heuristic Evaluation**

- Match between system and the real world - system should use user's daily language, with words, phrases and concepts familiar to the user, rather than any specific system-oriented terms.
  - Similar layout to real-life physical calendar
  - Red color to indicate deadlines and other cautionary instructions like delete and reset.
  - Green color to indicate instructions like save (success).
  - Star beside deadlines in date view to indicate the importance.

- Plus sign to add a new event.

21 **Heuristic Evaluation**

- User control and freedom - users often use functions by mistake and may need a clearly marked 'emergency exit' to leave the unwanted state without having to go through an extended process of reverting the change.
  - Different themes and views, allowing users to choose according to their preference (freedom).
  - Options to edit and make changes to deadlines (control).
  - Prompt message before saving allowing users to rectify the errors or to discard them completely (emergency exit).

22 **Heuristic Evaluation**

- Consistency and standards - users should have the experience of not having to wonder whether different words, situations, or actions mean the same thing.
  - Users can see the timetable of classes in 'timetable view' and 'specific date view', and in both the types of views consistency in color-coding different courses is maintained
  - Even the overall UI also maintains this color consistency.

23 **Heuristic Evaluation**

- Error prevention - even better than good error messages is a careful design that prevents a problem from occurring in the first place
  - A prompt is added for confirmation when there is any change in the view whether its change in background theme or addition or deletion of any events for any specific date, which helps in error prevention
  - Adding time while creating an event is handled by another view that is solely just designed to take correct time and correct format, thus minimizing human error

24 **Heuristic Evaluation**

- Recognition rather than recall - minimize user's memory load by making objects, actions, and options visible.
  - Adding events, saving events, deleting events and changing views all have well known and heavily used icons to recognize the feature
  - Using any feature in the application doesn't need recall for the user as the function names are quite self-explanatory
  - While creating an event, the user can see some text in input fields describing what should be entered in those input fields, which is an effective measure for recognition for the user

25 **Heuristic Evaluation**

- Flexibility and efficiency of use (for novice, intermittent, expert users)
  - Since the application is quite simple in nature, hence from novice to expert users all have the same simple interface

26 **Heuristic Evaluation**

- Aesthetic and minimal design - dialogues should not contain information that is irrelevant or rarely needed
  - Different types of views have the most appropriate amount of information that is required

- Simple UI in all types of views doesn't intimidate the users and hence restrict the user from abandoning the application

27 **Heuristic Evaluation**

- Help users recognize, diagnose, and recover from errors
  - The calendar allows users to delete any already set events along with an extra layer of protection prompt asking for confirmation; user can easily revert the changes back when needed.
  - While adding a deadline the view has a star along with it. If this is not the case then it helps the user to recognize that something might have gone wrong

28 **Heuristic Evaluation**

- Help and documentation
  - Since it mimics a basic calendar, most of the functionalities can be naturally understood by the user and no documentation is required as such.
  - Precise titles clearly indicate the functionality
  - For several functionalities (e.g. description), a small explanation is also provided in the white space.

29

Evaluation Results

30 **Heuristic Evaluation**

- Evaluation 1
  - Visibility of the system status:
    -
    - The system status is not visible for any instances. For example when I set a new event the system does not show any status (success or failure)
  - Match between system and real world:
    -
    - All the symbols/ words used in the system are matched to the real world. These are commonly used word/symbols

31 **Heuristic Evaluation**

- Evaluation 1
  - User Control and freedom:
    -
    - There are no undo and redo features for some actions
  - Consistency and standards:
    -
    - The system is not consistent with the various calendar applications. The deadline view, time table view and the schedule view buttons are placed in the places which is not common in the calendar apps. A separate menu bar should be provided to select these options

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32 **Heuristic Evaluation**

- Evaluation 1
  - Error Prevention:
    -
    - The save and delete buttons in the event creation page are far enough to select the proper option. When I try to delete a particular event or I try to reset the theme , a pop up comes up to confirm if I really want to do that
  - Recognition rather than recall:
    -
    - While creating an event, the I can see some text in input fields describing what should be entered in those input fields, which is an effective measure for recognition for the user
    -

33 **Heuristic Evaluation**

- Evaluation 1
  - Flexibility and efficiency of use:
    -
    - Since the application is quite simple in nature, hence from novice to expert users all have the same simple interface
  - Aesthetic and minimalist design:
    -
    - Only relevant details are shown in the system. No irrelevant details are asked or displayed in the system
    -

34 **Heuristic Evaluation**

- Evaluation 1
  - Help users recognize, diagnose and recover from errors:
    -
    - While adding an event if the event fails to add events, it does not show the reason. Making it difficult to diagnose it
    -
  - Help and documentation:
    -
    - No help and documentation is observed

35 **Heuristic Evaluation**

- Evaluation 2
  - Visibility of the system status:
    -
    - While adding event or uploading custom background the system state is not visible
  - Match between system and real world:
    -
    - Used red color to indicate deadlines and other cautionary instructions like delete and reset
    - Used green color to indicate instructions like save
    -

36 **Heuristic Evaluation**

- Evaluation 2
  - User Control and freedom:
    -
    - When delete option is selected a prompt pops up to undo it
    - Once event is created it can be edited
  - Consistency and standards:
    -
    - The menus are placed in the odd place
    -

37 **Heuristic Evaluation**

- Evaluation 2
  - Error Prevention:
    -
    - Pop ups appears to prevent errors
    - While entering date/time for the events date/time interface menu pops up to enter date time in correct format
  - Recognition rather than recall:
    -
    - Adding events, saving events, deleting events and changing views all have well known icons to recognise the feature
    -
    -

38 **Heuristic Evaluation**

- Evaluation 2

- Flexibility and efficiency of use:
  -
  - It is flexible for both novice and experienced users
- Aesthetic and minimalist design:
  -
  - There aren't any dialogues that contain information which is irrelevant or rarely needed
  -

39 **Heuristic Evaluation**

- Evaluation 2
  - Help users recognize, diagnose and recover from errors:
    -
    - While adding custom background if png image is selected the background is not loaded and the no information is displayed for not applying the change, making it hard to diagnose
    - No information(reason) is displayed when event a event creation is failed
    -
  - Help and documentation:
    -
    - No help and documentation is found

40 **Heuristic Evaluation**

- Evaluation 3
  - Visibility of the system status:
    -
    - Does not show any system status while creating/editing/deleting events
  - Match between system and real world:
    -
    - Add/ edit/ delete icons are used as it is used in other applications
    -

41 **Heuristic Evaluation**

- Evaluation 3
  - User Control and freedom:
    -
    - Users have the control over the created events
  - Consistency and standards:
    -
    - The set background button looks bigger than other buttons

42 **Heuristic Evaluation**

- Evaluation 3
  - Error Prevention:
    -
    - Used ups to prevent cautionary functions like delete, reset and replace event data (edit)
  - Recognition rather than recall:
    -
    - The functionalities could be recognised by the way it is presented
    -

43 **Heuristic Evaluation**

- Evaluation 3
  - Flexibility and efficiency of use:
    -
    - The system is very simple and can be used by both novice and expert users
  - Aesthetic and minimalist design:
    -
    - Only relevant details are given while viewing dates, events and creating new events

44 **Heuristic Evaluation**

- Evaluation 3
  - Help users recognize, diagnose and recover from errors:
    -
    - While creating event if some information is not given the event is not created. However, this reason is not displayed to the user
    -
  - Help and documentation:
    -
    - No help and documentation there

45 **Heuristic Evaluation - Summary**

46 **Conclusion**

- Evaluators noticed that no system state was visible. A message can be shown once a event is created/edited/deleted and background is applied
- Evaluator 1 pointed out the need of undo and redo. Undo and redo operation can be incorporated by adding undo/redo for the last activity. However, we have edit option

for created events

- According to the evaluators, the buttons for the view are not expected in the front page. This can be added in a separate menu bar
- While creating events and setting backgrounds the reason of any error can be displayed as a error message
- Since it mimics a basic calendar, most of the functionalities can be naturally understood by the user and no documentation is required as such
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  -

47 **Thank You**